

AERO50009 Propulsion and Turbomachinery Tutorial Sheet 1 (version of January 28, 2024)

In this tutorial you will use two of the tools available at

[AERO50009 Propulsion and Turbomachinery web tools at Imperial College](#)

(The web page gives the year as 2023, but it is the same for 2024.)

Click on, have a look, and add the link to your favourites for future use.

You will need to make reasonable (and easy) guesses when using the tools, generally developed for real engines, when using them for ideal engines. This should be OK, and if unsure, discuss with other students first.

1. Sketch a turbojet h - s diagram, and annotate it with the meaning of the enthalpy changes such as the flight velocity etc. Then figure out which elements or combination of elements correspond to the following parameters: specific thrust, propulsion efficiency, and the thermal efficiency. Then add to this thrust-specific fuel consumption and overall efficiency.

2. Sketch the h - s diagram of an ideal turbojet engine for take-off conditions ($u = 0$). Compare such diagrams for three engines, each designed to provide the maximum possible take-off thrust. Engine 1 has a very small compressor pressure ratio, Engine 2 has a medium compressor ratio, and Engine 3 has a large compressor ratio.

Use the the h - s diagram comparison tool to verify your sketches.

Now, using the diagram (that is neglecting the small effect of mass addition of fuel in the burner, which the diagram does not reflect, find, for a given maximum enthalpy $h_{\max}$ and ambient enthalpy h_a , the compressor pressure ratio such that the engine will have a maximum take-off thrust.

In the following question assume that all specific heat ratios are equal to 1.4 and that all gas constants are equal to 287 J/(kg K).

3. Calculate the fuel-to-air ratio, specific thrust, and overall efficiency of an ideal ramjet engine for the following conditions:

Flight Mach number: 3.

Burner output temperature: 1600K.

Fuel heat of reaction: 45000 kJ/kg.

Specific heat at constant pressure for the combustion products 1005 J/(kg K).

Ambient temperature 215K (corresponds to the altitude of 12000m).

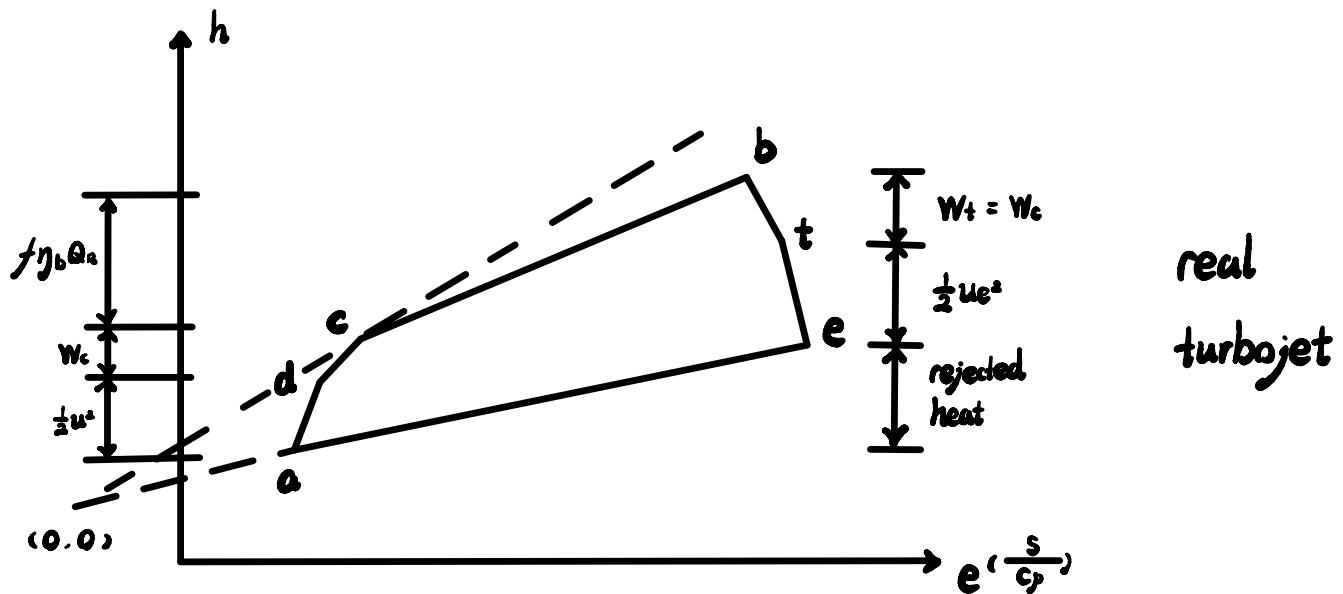
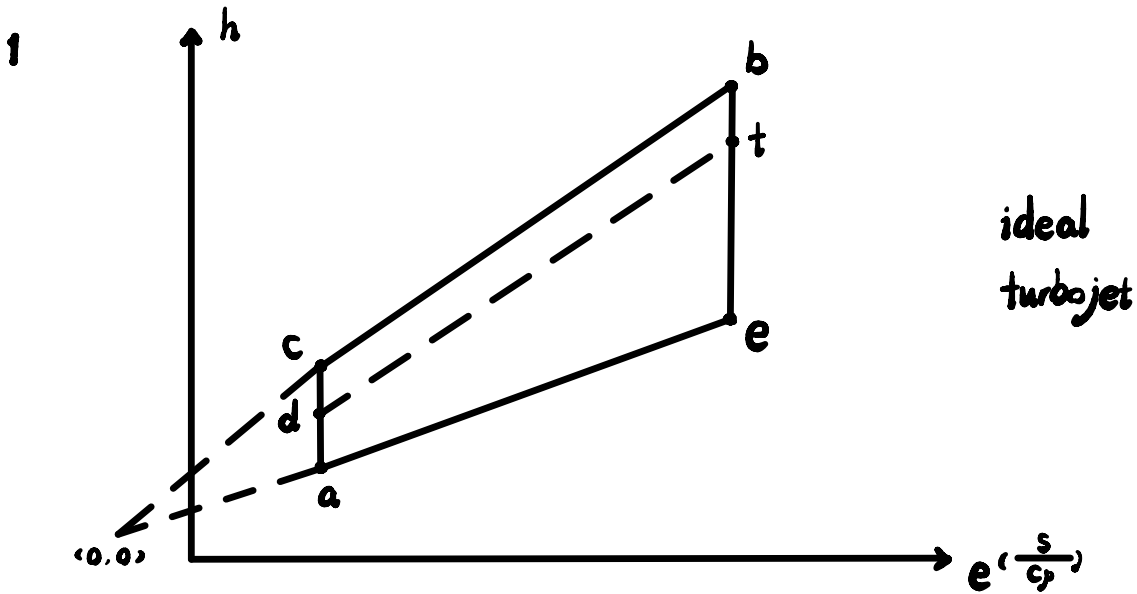
Check your answers using the turbofan calculator as described below.

Important! Use Turbofan Calculator to check your calculations step-by-step. To model ramjet as a turbofan, prescribe the compressor pressure ratio $p_r = 1$ and bypass ratio $\beta = 0$. Give reasonable values to other input parameters of the fan, with $\beta = 0$ and $p_r = 1$ they will not affect the results.

Pressing the ramjet button and then the "Make engine ideal" fill in the right values for this tutorial. You need, however, to learn to calculate it on your own.

Note that if you come to the lecturer saying that you cannot get the correct answer in this tutorial, the lecturer will ask "At what step your result deviates from that of the online turbofan calculator?" If you will not be able to answer, you will be asked to go and work with the online calculator first.

Do this on your own at home before the in-class tutorial session. The tutorial session is a discussion of the work you do in the tutorial, driven by your questions.



$$\frac{F}{\dot{m}_a} = (1+f) u_e - u \approx u_e - u$$

$$\eta_{prop} = \frac{F u}{\frac{1}{2} \dot{m}_a [(1+f) u_e^2 - u^2]} \approx \frac{(u_e - u) u}{\frac{1}{2} (u_e^2 - u^2)} = \frac{2u}{u_e + u}$$

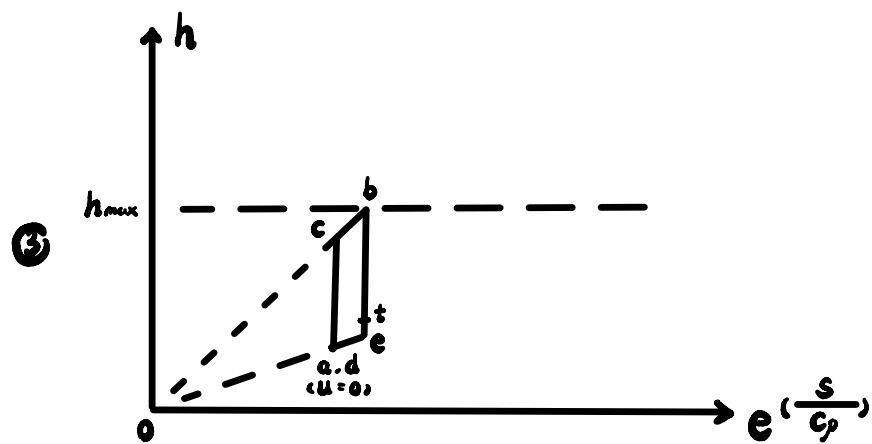
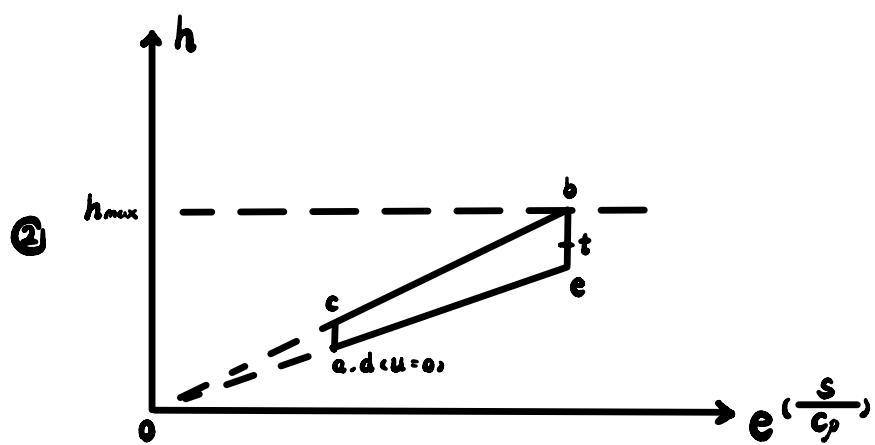
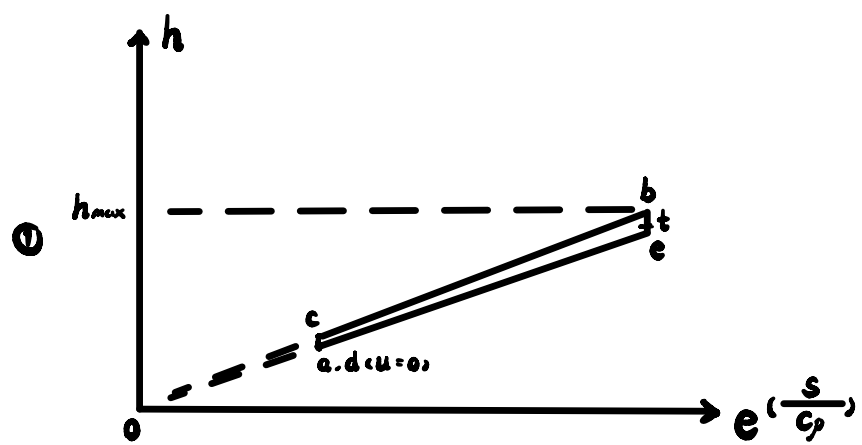
$$\eta_{th} = \frac{\frac{1}{2} \dot{m}_a [(1+f) u_e^2 - u^2]}{\dot{m}_a f Q_R} \approx \frac{\frac{1}{2} u_e^2 - \frac{1}{2} u^2}{f Q_R} = \frac{f Q_R - \text{rejected heat}}{f Q_R} = 1 - \frac{\text{rejected heat}}{f Q_R}$$

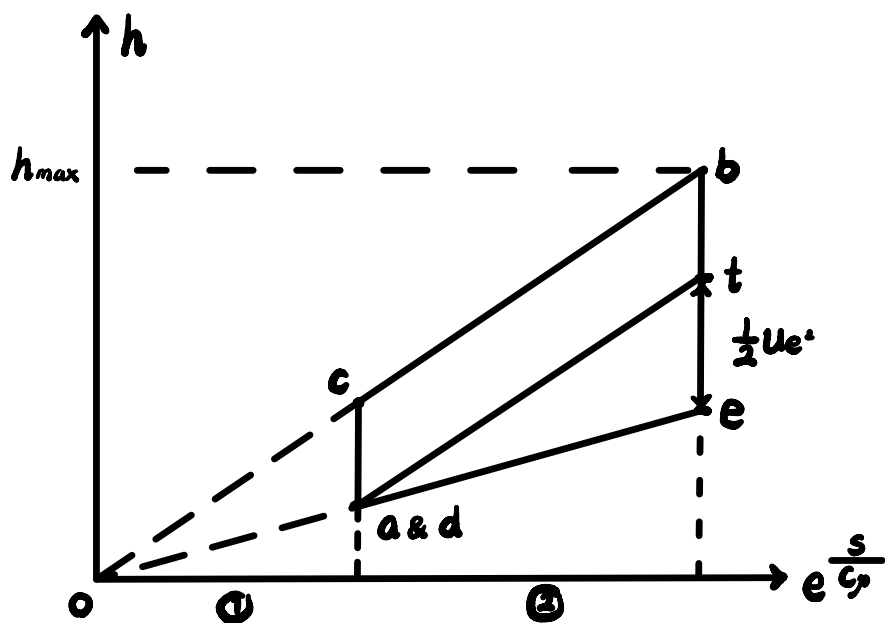
$$\text{for ideal engine, } \eta_{th} = 1 - \frac{T_a}{T_c} = 1 - \frac{T_e}{T_b}$$

$$\eta_o = \eta_{prop} \eta_{th}$$

$$TSFC = \frac{\dot{m}_f}{F} = \frac{\dot{m}_f}{\dot{m}_a [(1+f) u_e - u]} \approx \frac{f}{u_e - u}$$

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$f \ll 1$. therefore for ideal turbojet

$$F = \dot{m}_a (u_e - u) = \dot{m}_a u_e \text{ for } u = 0$$

$$\therefore \frac{1}{2} u e^2 = h_{ot} - h_e, \quad p_{rc} = \frac{p_{oc}}{p_a} = \left(\frac{T_{oc}}{T_a} \right)^{\frac{\delta-1}{\delta}} = \left(\frac{h_{oc}}{h_u} \right)^{\frac{\delta-1}{\delta}}$$

where $h_{\max} - h_{\text{ot}} = h_{\text{oc}} - h_a \quad \therefore h_{\text{ot}} = (h_{\max} + h_a) - h_{\text{oc}}$

according to diagram, $\frac{h_a}{h_e} = \frac{h_{oc}}{h_{max}} = \frac{①}{① + ②}$

$$\therefore \frac{1}{2} u e^2 = h_{\max} + h_u - h_{oc} - h_e = h_{\max} + h_u - h_{oc} - \frac{h_u}{h_{\max} h_{oc}}$$

$$\therefore \frac{d(\frac{1}{2}ue^*)}{dh_{oc}} = -1 + \left(\frac{h_u}{h_{max}}\right) \frac{1}{h_{oc}^2} = 0 \text{ for maximum } \frac{1}{2}ue^*$$

$$\therefore h_{rc} = \sqrt{\frac{h_u}{h_{max}}} \quad \therefore p_{rc} = \left(\frac{1}{\sqrt{h_u h_{max}}} \right)^{\frac{8-1}{8}}$$

$$\therefore \frac{1}{2} u_e^2 = h_{ot} - h_e = h_{max} - (h_{max} - h_{ot}) - h_e$$

$$= h_{max} - (h_{oc} - h_a) - h_e$$

$$\text{where } \frac{h_{oc}}{h_a} = p_{rc}^{\frac{\delta-1}{\delta}} \quad \therefore h_{oc} = p_{rc}^{\frac{\delta-1}{\delta}} h_a$$

$$\text{also, according to the diagram, } \frac{h_{max}}{h_{oc}} = \frac{h_e}{h_a}$$

$$\therefore h_e = \frac{h_{max}}{h_{oc}} h_a = h_{max} p_{rc}^{-\frac{\delta-1}{\delta}}$$

$$\therefore \frac{1}{2} u_e^2 = h_{max} - p_{rc}^{\frac{\delta-1}{\delta}} h_a + h_a - h_{max} p_{rc}^{-\frac{\delta-1}{\delta}}$$

$$\therefore \frac{d(\frac{1}{2} u_e^2)}{dp_{rc}} = -\frac{\delta}{\delta-1} p_{rc}^{(\frac{\delta-1}{\delta})-1} h_a + \frac{\delta-1}{\delta} h_{max} p_{rc}^{(-\frac{\delta-1}{\delta})-1} = 0$$

$$\therefore h_a p_{rc}^{\frac{\delta-1}{\delta}} = h_{max} p_{rc}^{-(\frac{\delta-1}{\delta})}$$

$$\therefore p_{rc}^{2(\frac{\delta-1}{\delta})} = \frac{h_{max}}{h_a}$$

$$\therefore p_{rc} = \left(\frac{h_{max}}{h_a} \right)^{\frac{\delta}{2(\delta-1)}}$$

$$3 \quad C_p = 1005 \text{ J/kgK}, \quad \gamma = 1.4 \quad \text{and} \quad R = 287 \text{ J/kgK}$$

$$\therefore f Q_R + h_{od} = (1+f) h_{ob}$$

$$\therefore f = \frac{h_{ob} - h_{od}}{Q_R - h_{ob}} = \frac{T_{ob} - T_{od}}{Q_R/C_p - T_{ob}}$$

$$T_{od} = T_a \left(1 + \frac{\gamma-1}{2} M^2\right) = 2.8 T_a = 602 \text{ (K)}$$

$$\therefore f = 0.02311$$

$$u = M a = M \sqrt{\gamma R T_a} = 881.75 \text{ (m/s)}$$

$$h_{oe} = h_e + \frac{1}{2} u_e^2 \quad \therefore u_e = \sqrt{2(h_{oe} - h_e)}$$

$$h_{oe} = h_{ob} = C_p T_{ob} = 1608000 \text{ (J/kg)}$$

$$h_e = C_p T_e$$

$$\therefore \frac{T_e}{T_{ob}} = \frac{T_e}{T_{ob}} = \left(\frac{p_a}{p_{ob}}\right)^{\frac{\gamma-1}{\gamma}} = \left(\left(\frac{T_a}{T_{ob}}\right)^{\frac{\gamma}{\gamma-1}}\right)^{\frac{\gamma-1}{\gamma}} = \frac{5}{14}$$

$$\therefore h_e = \frac{5}{14} C_p T_{ob} = 574285.7143 \text{ (J/kg)}$$

$$\therefore u_e = 1437.856 \text{ (m/s)}$$

$$\therefore \frac{F}{m_a} = (1+f) u_e - u = 589.33 \text{ (Ns/kg)}$$

$$\therefore \eta_o = \frac{F u}{m_a Q_R} = \frac{[(1+f) u_e - u] u}{f Q_R} = 0.497$$